

Abdul Rafi

Computer Science & Engineering Undergraduate

Robotics & Autonomous Systems • Artificial Intelligence • Computer Vision • Embedded Systems

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University of Moratuwa, Sri Lanka

Profile

Computer Science and Engineering undergraduate at the University of Moratuwa with a strong interest in robotics, autonomous systems, artificial intelligence, computer vision, and embedded systems. I enjoy working across both hardware and software, with experience in humanoid and mobile robotics, ROS 2 navigation and SLAM, and full-stack system integration. My main focus is on bringing perception, control, and software together to build intelligent systems that can work reliably in real-world environments.

Education

University of Moratuwa, Sri Lanka March 2024 – Present

B.Sc. Engineering (Hons.) in Computer Science & Engineering

Specialization in Integrated Computer Engineering

CGPA: 3.15 / 4.00

St. Thomas' College, Matara

January 2009 – January 2023

G.C.E. Advanced Level (Physical Science Stream)

- Achieved 3 A grades (Combined Mathematics, Physics, Chemistry)
- All-Island Rank: 206 | Z-Score: 2.5139

Engineering Projects

Unitree G1 Humanoid Robot – Localization & Navigation

🐙 [GitHub](#) • 📄 [Report](#)

Unitree G1 | ROS 2 | Humanoid Robotics | Localization | LLM + RAG

- Working with the Unitree G1 humanoid robot in simulation on robotic localization and navigation, integrating perception, sensing, and autonomous behaviors.
- Implemented ROS 2-based sensor integration and simulated robotic experimentation to validate autonomous navigation performance.
- Integrated an LLM-powered Retrieval-Augmented Generation (RAG) system to ground model responses in retrieved knowledge.

ROS 2 Autonomous Bathymetric Survey Drone

🐙 [GitHub](#) • 📄 [Report](#)

ROS 2 | ArduPilot | Computer Vision | Autonomy

- Built an autonomous reservoir-survey drone using ROS 2 Jazzy, ArduPilot SITL, and Gazebo Harmonic for mission planning and waypoint navigation.
- Implemented HSV-based computer vision for water detection combined with sonar depth acquisition and a state-machine-driven autonomous mission.
- Developed a React-based Ground Control Station with Three.js bathymetry visualization and real-time telemetry.

QBot Autonomous Navigation System

🐙 [GitHub](#) • 📄 [Report](#)

ROS 2 | SLAM | LiDAR | Nav2

- Developed indoor mapping and navigation using ROS 2, LD19 LiDAR, and SLAM Toolbox.
- Implemented A* path planning with Pure Pursuit control for autonomous goal navigation.
- Built a React dashboard for live map visualization and manual robot control.

Nano Processor

🐙 [GitHub](#)

VHDL | FPGA | Digital Systems

- Designed and implemented a processor architecture in VHDL as a digital systems engineering project.
- Verified system-level digital logic and processor functionality on FPGA hardware.

School Management System



Real Client Project | React | FastAPI | Supabase | PostgreSQL | Mobile + Push Notifications

- Built a full-stack attendance and student management platform (React + TypeScript, React Native, FastAPI, Supabase/PostgreSQL) for a school of approximately 700 students, used in production by real teachers and staff.
- Implemented role-based access, scheduled push notifications, and automated Excel/PDF reporting; owned the project end-to-end from requirements to deployment.

Technical Skills

Programming Languages – Python, Java, JavaScript, TypeScript, C++, C

Robotics & Autonomous Systems – ROS 2, Nav2, SLAM, LiDAR, ArduPilot, MAVLink, Gazebo, MuJoCo

AI & Computer Vision (Working Knowledge) – OpenCV, LangChain, LLMs, RAG, YOLO

Embedded Systems & Hardware – STM32, Arduino, ESP32, FPGA (VHDL)

Web Development – React, React Native, Node.js, FastAPI

Databases – MySQL, PostgreSQL, MongoDB

Engineering Tools – Git, Docker, CMake, Linux (Ubuntu)

Publications

- **M.A.A. Rafi** and S. Sooriyaarachchi, “Intelligent Humanoid Airport Concierge: A RAG-Based Voice Interface and Autonomous Navigation System for the Unitree G1,” submitted to *ERU Symposium 2026*, University of Moratuwa, Sri Lanka. (*Under Review*)

Leadership & Service

- Programme Committee Member – Hit the Ground Tournament, University of Moratuwa
- Logistics Committee Member – SLIoT Competition, University of Moratuwa
- Logistics & Marketing Team Member – Career Fair 26
- Religious Coordinator – Majlis-Ul-Islam, University of Moratuwa

Achievements

- Runner-up – IEEE INSL Idea Stage 2025 Competition
- Top 50 – SPARK Competition – ROS 2 Autonomous Bathymetric Survey System with ArduPilot
- Finalist – YARL Geek Challenge Season 14
- Finalist – Sri Lanka Robotics Challenge (SLRC) 2025

References

Prof. Chandana D. Gamage

Professor

University of Moratuwa, Sri Lanka

B.Sc. Eng. (Moratuwa), M.Eng. (AIT),

Ph.D. (Monash), CEng, MIE(SL), MACM

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Dr. S. J. Sooriyaarachchi

Senior Lecturer

University of Moratuwa, Sri Lanka

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